

QBE_{max} is a sequence-permuted and internally protected base editor

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Jiacheng Hu¹, Mengyue Guo¹, Qiang Gao¹, He Jia¹, Mingyang He¹,
Zhiwei Wang¹, Lina Guo¹, Guanwen Liu^{1,2}, Quan Gao¹ &
Kevin Tianmeng Zhao¹✉

Cytosine base editors (CBEs) show promise for multiplex gene knockout applications, but impure edits, indels and off-targets still frequently occur. We describe here QBE_{max}, which exhibits high efficiency, low indel and off-targets and high product purity with up to 99.8% of edits comprised of C-to-T. Through molecular dynamic modeling, QBE_{max} presents as a compact and stable base editor that shields protected bases from undesired repair processes.

The two most common classes of base editors are modular fusion proteins comprised of a deaminase and DNA-binding protein to perform targeted and precise C·G-to-T·A (cytosine base editors (CBEs)) or A·T-to-G·C (adenine base editors (ABEs)) base conversions with high efficiency^{1,2}. In addition to the use of base editors to correct disease-causing genetic mutations, base editors have shown promise as an alternative to nucleases for multiplex gene knockout applications. CBEs can precisely edit arginine, glutamine or tryptophan codons to generate premature stop codons; because this process does not undergo the formation of DNA double-strand breaks, base editing for multiplex gene knockouts minimizes the risk of genomic translocations, cell toxicity and DNA chromothripsis^{3,4}. However, unintended indels and impure base editing byproducts (for example, C-to-G and C-to-A) are still frequently observed when using CBEs. The formation of impure edits could be detrimental and act as a missense mutation in a gene otherwise targeted for knockout. In this regard, an ideal CBE for gene knockout should exhibit properties such as high efficiency, low indel formation, low off-target edits, an expanded editing window to reach more potential bases and, importantly, high product purity for safety purposes.

Because canonical base editors are comprised of different proteins fused together end-to-end, each protein's orientation may not be best to balance catalytic processes including nontarget strand association/dissociation by the deaminase, target base deamination, base protection/exposure and, ultimately, endogenous cellular DNA repair enzymes. Previous engineering studies have demonstrated that either the use of an inlaid deaminase domain or a circularly permuted Cas protein could affect the on-target or off-target editing efficiencies, product purity or editing windows of base editors^{5–9}. However, the

combination of these parameters has not been optimized using any one approach. To obtain a more ideal base editor, we hypothesized that we could treat the deaminase and Cas9 protein together as one complex comprised of different domains. Therefore, by shuffling the orientation of domains from both the deaminase and Cas9 together, which combines concepts of circularly permuted proteins and inlaid deaminases, we hoped to identify a CBE that maximizes the beneficial properties of using CBEs for gene knockout.

We first designed four base editor orientations in which a deaminase was internally embedded within a circularly permuted Cas9(D10A) protein¹⁰ (Fig. 1a) based on Cas9 positions previously found to be amenable for circular permutation or for inlaying a small peptide^{10,11}. These architectures are hereby designated as Q base editors (QBE1 to QBE4) to reflect the reconstitution process of circularizing domains from the Cas9 protein and inserting a deaminase internally to generate a new start codon (circle with an internal cut resembling the letter 'Q'). To characterize the properties of different QBE architectures, six different cytidine deaminases (rAPOBEC1 (ref. 1), hA3A, mini-Sdd3, mini-Sdd6, Sdd7 and mini-Sdd9 (ref. 12)) were evaluated.

Plasmids encoding corresponding CBEs were transfected in HEK293T cells and compared with the canonical BE4_{max} architecture¹³ at two endogenous genomic sites. Deep sequencing revealed that QBE3-based editors showed comparable or higher editing frequencies for four of the six deaminases evaluated; in contrast, QBE1-based, QBE2-based and QBE4-based editors exhibited lower editing frequencies (Supplementary Fig. 1a,b). Compared to the BE4_{max} counterparts, the QBE3-based editors showed substantially lower indels, with an average indel reduction of 60.4% for site 1 and 62.6% for site 3 (Supplementary Fig. 1c,d). We next measured the edit-to-indel ratios and

¹Qi Biodesign, Beijing, China. ²Center for Genome Editing, Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Beijing, China.

✉e-mail: kzhao@qi-biodesign.com

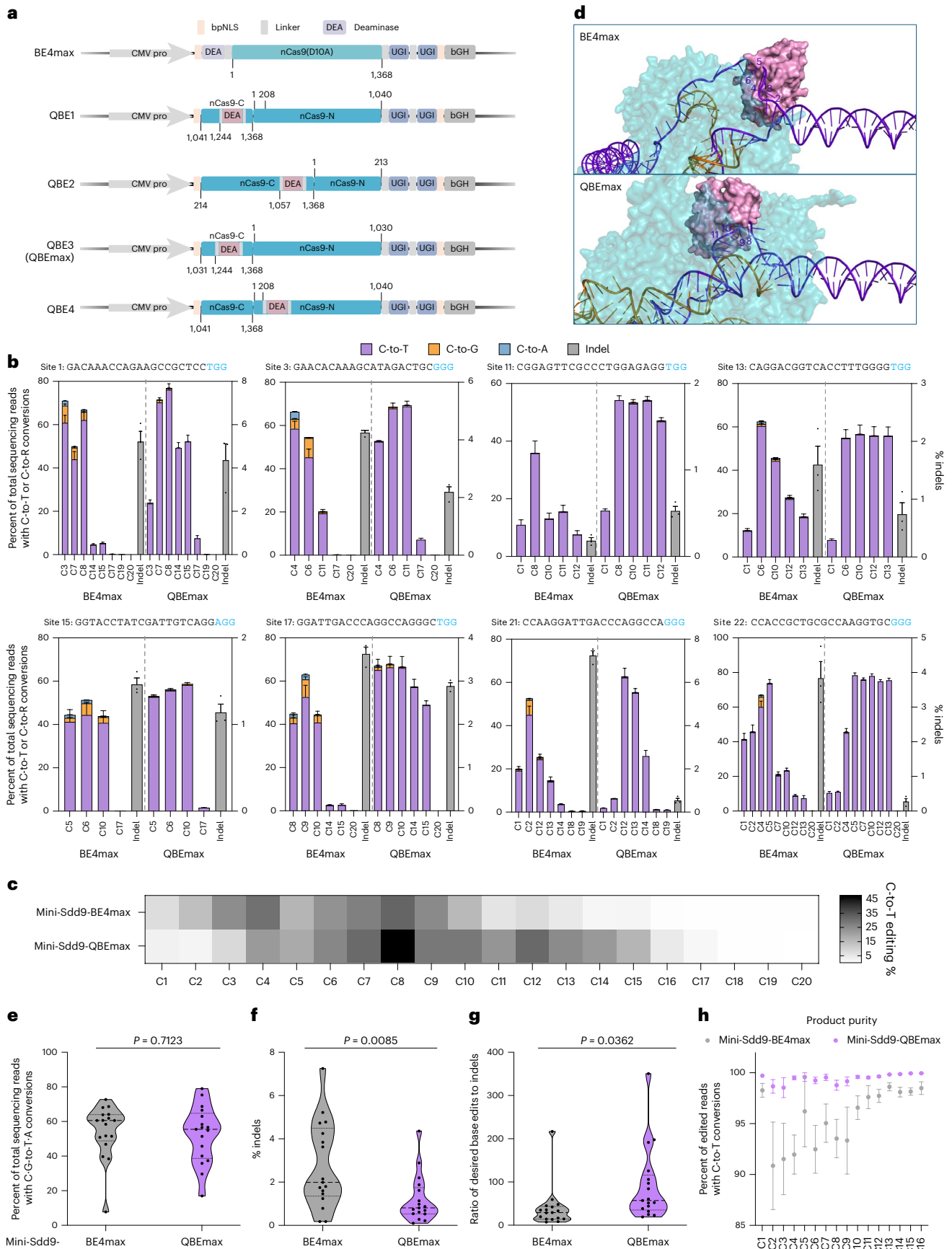


Fig. 1 | Design and characterization of QBEmax. **a**, Schematic representations of the BE4max, QBE1, QBE2, QBE3 (QBEmax), QBE4 editors. Numbers under and above nCas9(D10A) represent amino acid positions in reference to wild-type SpCas9. **b**, Frequencies of C-to-T or C-to-R conversions (left y axis) and indels (right y axis) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax; values and error bars represent the means and s.e.m. for three independent biological replicates. **c**, Gray scale heat map showing average cytosine base editing frequencies by mini-Sdd9-BE4max and mini-Sdd9-QBEmax at each protospacer position across 17 endogenous sites tested. Numbers below the heat map indicate protospacer positions. **d**, AlphaFold3-predicted structures of mini-Sdd9-BE4max and mini-Sdd9-QBEmax bind to a sgRNA and DNA target (site 1). Blue, Cas9 domains; orange, sgRNA; purple, dsDNA target; pink, mini-

Sdd9. **e–g**, Average editing frequencies of C-to-T conversions (**e**), indels (**f**) and ratio of base edit-to-indel (**g**) induced by the mini-Sdd9-BE4max or mini-Sdd9-QBEmax editors across 17 endogenous sites; each dot represents the mean for three independent biological replicates for a specified target site, the violin plot shows the base editing frequency distribution with medians and quartiles, and significances are indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBEmax by exact *P* value using two-tailed Student's *t*-test, *n* = 17. **h**, Percent of edited reads with C-to-T conversions among edited events at each C1 to C16 base position, cumulated across 17 endogenous sites; values and error bars represent the means and s.e.m. for three independent biological replicates. CMV pro, enhanced cytomegalovirus promoter; DEA, deaminase; NLS, nuclear localization signals; bGH, bovine growth hormone polyadenylation signal.

found that six of the seven editors at site 1, and five of the seven editors at site 3 showed substantially higher edit/indel ratios (Supplementary Fig. 1e,f). We then analyzed the editing window and product purities of QBE editors. We observed higher editing efficiencies at PAM-proximal Cs for QBE3 editors and greatly improved product purities compared to BE4max editors (Supplementary Fig. 2a,b). Based on these, we hereby refer to QBE3 as QBEmax (Fig. 1a).

From initial evaluations, we found that mini-Sdd9-based QBE editors exhibited superior editing properties in terms of editing activity, indel formation and product purity. Using mini-Sdd9, we designed seven additional QBEs (QBE5–QBE11; Supplementary Fig. 3a). We found that only one editor, QBE6 demonstrated similar performance to QBEmax in terms of editing efficiency and purity; however, its editing window appeared narrow, so we hereby designate it as QBE_n (Supplementary Fig. 3b,c). Because of mini-Sdd9-QBEmax's overall superior performance and relatively wide editing window, which is desired for expanding the targeting scope of gene knockout applications with base editors, we selected it for further study.

It was reported that fusing the deaminase to the N terminus of a circularly permuted Cas9 (referred to as CP-BE)⁵ could broaden the editing window, and inlaying the deaminase within the Cas9 protein (referred to as inlaid-BE)^{6–9} could affect the editing efficiency or off-target efficiency. We next compared mini-Sdd9-QBEmax with CP-BEs and inlaid-BEs with Cas9 permutation or deaminases inlaid at positions used in the QBE1–QBE11 editors. We found that all CP-BEs and inlaid-BEs induced more impure products than QBEmax. Notably, QBEmax (which uses positions 1,031 and 1,244 for modular assembly) outperforms individual CP-1031 and inlaid-1244 CBEs when comparing the combination of editing frequencies, indel formation and product purities (Extended Data Fig. 1), suggesting that the modularly designed QBEmax architecture enhances desired properties from each functional domain.

To further profile editing properties of mini-Sdd9-QBEmax, we compared mini-Sdd9-QBEmax with mini-Sdd9-BE4max across 17 endogenous genomic sites in HEK293T cells. We found that, in contrast to mini-Sdd9-BE4max, which biases editing at the PAM-distal

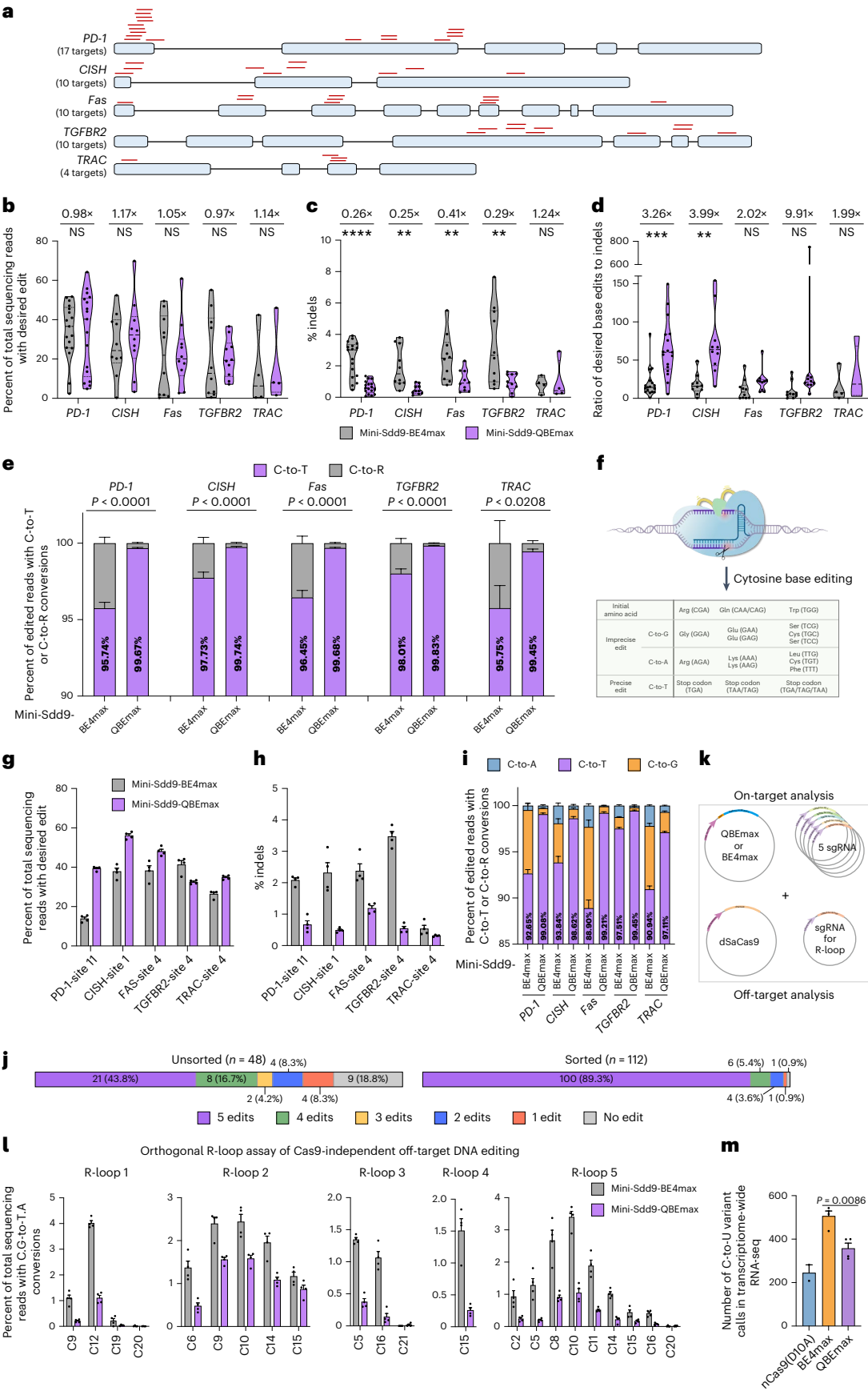
region, mini-Sdd9-QBEmax showed a wider editing window as far as C16 (Fig. 1b and Supplementary Fig. 4). Aggregate analyses revealed a 'forward-shifted' and wider editing window for mini-Sdd9-QBEmax with target Cs between 4 and 14 being favored (Fig. 1c). We used AlphaFold3 (ref. 14) to predict a mini-Sdd9-QBEmax–sgRNA–target DNA ternary structure and compared it with that of the corresponding BE4max architecture (Fig. 1d). We found that these structures revealed the deaminase in mini-Sdd9-QBEmax being more closely associated to PAM-proximal Cs, while in BE4max, more closely associated to PAM-distal Cs, which is consistent with experimental results from genomic edits.

The average editing frequencies across 17 genomic sites for mini-Sdd9-QBEmax and mini-Sdd9-BE4max were $52.4 \pm 2.4\%$ and $54.5 \pm 2.2\%$, respectively (Fig. 1e). Mini-Sdd9-QBEmax induced lower indels at 16 of 17 sites tested, with average indel frequencies decreasing by 56.5% from $2.8 \pm 0.3\%$ to $1.2 \pm 0.2\%$ (Fig. 1f and Supplementary Fig. 4), which also substantially increases average edit-to-indel ratios (Fig. 1g). We next evaluated cytosine base editing product purities, which is calculated as the proportion of 'C' edited to 'T' as opposed to 'G' or 'A', for each position within the protospacer and aggregated all sites together. Importantly, mini-Sdd9-QBEmax exhibited superior product purities ($99.4\% \pm 0.4\%$) at all positions within a C1–C16 editing window compared to that of mini-Sdd9-BE4max ($95.5\% \pm 2.8\%$; Fig. 1b,h and Supplementary Fig. 4). To test the versatility of the QBEmax, we evaluated mini-Sdd9-QBEmax in additional mammalian cell lines, including A549, HeLa and HCT116. QBEmax exhibited higher or comparable editing efficiencies, improved product purities and decreased indels in all cell lines tested (Extended Data Fig. 2). These results highlight QBEmax in achieving efficient and precise base edits at a flexible editing window with minimal indel and byproducts.

Chimeric antigen receptor-T cell (CAR-T) therapy has demonstrated success as a cancer immunotherapy for hematological malignancies. Many clinical trials and research studies have found that multiplex knockout of genes related to immune rejection and graft-versus-host disease (GvHD) would further benefit CAR-T therapy in terms of both durability and potency. Encouraged by the

Fig. 2 | Multiplex base editing gene knockout of CAR-T targets. **a**, Schematic representations showing *PD-1*, *CISH*, *Fas*, *TGFB2* and *TRAC* genes. Light blue boxes indicate exons of genes, and short red lines represent the position of selected protospacers. **b–d**, Average editing frequencies of desired editing efficiencies (**b**), indels (**c**) and ratio of base edits to indels (**d**) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax editors for *PD-1* (*n* = 17), *CISH* (*n* = 10), *Fas* (*n* = 10), *TGFB2* (*n* = 10) and *TRAC* (*n* = 4) genes; each dot represents the mean for three independent biological replicates for a specified target site and the violin plot shows base editing frequency distribution with medians and quartiles. **e**, Percent of edited reads with C-to-T or C-to-R conversions at target genes indicated with values and error bars representing the means and s.e.m. for three independent biological replicates across all target sites for *PD-1* (*n* = 17), *CISH* (*n* = 10), *Fas* (*n* = 10), *TGFB2* (*n* = 10) and *TRAC* (*n* = 4) genes; significances are indicated between BE4max and QBEmax by exact *P* value using two-tailed Student's *t*-test. **f**, Schematic representations of potential editing outcomes

induced by C-to-T, C-to-G and C-to-A conversions of tryptophan (W), arginine (R) and glutamine (Q). **g–i**, Desired editing efficiencies (**g**), indels (**h**) and percent of edited reads with C-to-T or C-to-R conversions at target genes indicated (**i**) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax editors during multiplexed base editing of five genes; values and error bars represent the means and s.e.m., respectively, for four independent biological replicates. **j**, Single-cell colony analysis of multiplex base editing distributions in unsorted and sorted cell populations. **k**, Schematic representation of the experimental design for the R-loop assay. **l**, Frequencies of C-to-T conversions at the dSaCas9-induced R-loop sites; values and error bars represent the means and s.e.m. for four independent biological replicates. **m**, Number of C-to-U RNA variants induced by the editors indicated; values and error bars represent the means and s.e.m. for two (Cas9(D10A)) or four (BE4max and QBEmax) independent biological replicates, significances are indicated by exact *P* value using one-way ANOVA Tukey's multiple comparisons.



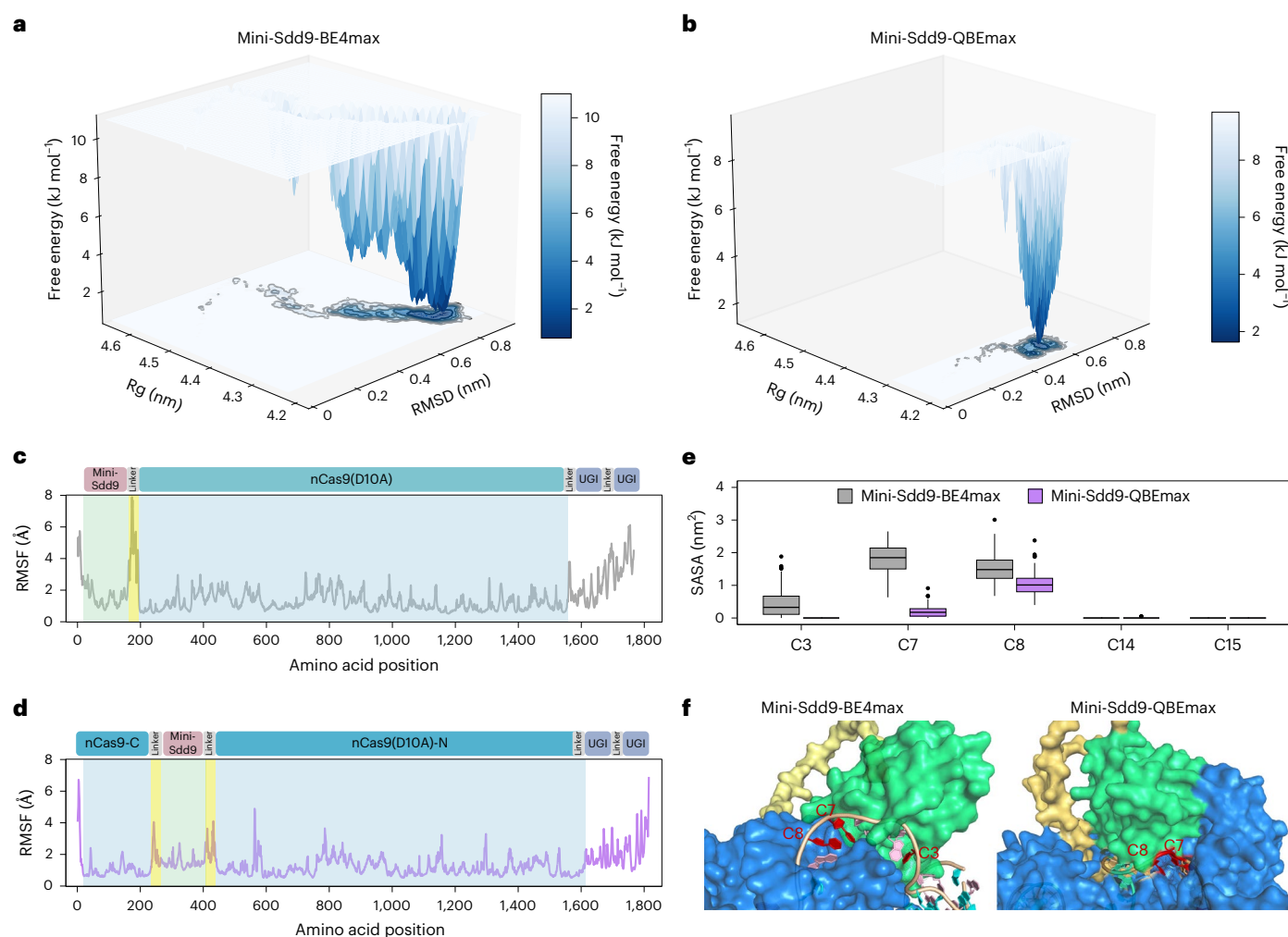


Fig. 3 | Molecular dynamic modeling reveals QBEmax as a more compact and protective editor. a,b, The free energy landscape against RMSD and Rg for mini-Sdd9-BE4max (**a**) and mini-Sdd9-QBEmax (**b**) during a 300 ns MD simulation. **c,d,** RMSF plot for mini-Sdd9-BE4max (**c**) and mini-Sdd9-QBEmax (**d**) in the MD simulation, systems equilibrated after 150 ns; schematic representations of editors are shown above the plots. **e,** SASA analysis of Cs within the editing

window of site 1 in predicted mini-Sdd9-BE4max and mini-Sdd9-QBEmax ternary structures; each replicate represents the SASA by a 1.0 nm probe during a 1 ns time scale; $n = 150$ ns following system equilibration, boxes and lines represent the interquartile range (IQR) and median, respectively, and whiskers represent $1.5 \times \text{IQR}$. **f,** Snapshots showing exposed Cs in the editing window. Blue, Cas9 and UGI; yellow, linker; green, mini-Sdd9 deaminase.

performance of mini-Sdd9-QBEmax, we next sought to perform multiplex gene knockout to simultaneously edit genes that could compromise the efficacy of CAR-Ts. Five genes, *PD-1* (ref. 15), *CISH*¹⁶, *Fas*¹⁷, *TGFBR2* (truncating off the endodomain)^{18–20} and *TRAC*²¹, were selected, which all previously demonstrated potential in improving CAR-T performance when knocked out or downregulated.

We first identified all possible SpCas9 protospacers with an NGG PAM and target C located within codons encoding tryptophan (W), arginine (R) or glutamine (Q), so that a cytosine base edit would generate a stop codon (TAA, TAG or TGA). We obtained 46, 21, 16, 30 and 4 protospacers for *PD-1*, *CISH*, *Fas*, *TGFBR2* and *TRAC*, respectively (Supplementary Table 1). We next filtered for potential off-target sites (mismatches ≤ 3) and ultimately selected 16 targets for *PD-1*, 9 for *CISH*, 10 for *Fas*, 10 for *TGFBR2* and 3 for *TRAC*. Lastly, we included one additional target for *CISH*, *PD-1* and *TRAC*, which disrupts a splice site to perform gene knockout as reported previously^{21,22} (Fig. 2a).

HEK293T cells were transfected with mini-Sdd9-BE4max or mini-Sdd9-QBEmax together with each sgRNA plasmid. We then analyzed desired editing efficiencies (calculated as percent C-to-T for stop codon creation), indel frequencies, desired edit-to-indel ratios and product purities. We found that mini-Sdd9-QBEmax achieved comparable or slightly higher average desired editing at the target

base for all sites aggregated for each of the five genes (Fig. 2b). Average indel frequencies for all sites aggregated decreased by 76%, 75%, 59% and 71% for *PD-1*, *CISH*, *Fas* and *TGFBR2*, respectively (Fig. 2c). The average indel frequency at *TRAC* was $0.8 \pm 0.15\%$ for mini-Sdd9-BE4max and $1.0 \pm 0.34\%$ for mini-Sdd9-QBEmax due to one outlier at *TRAC*-site 3 whereby mini-Sdd9-BE4max and mini-Sdd9-QBEmax exhibited $1.4 \pm 0.18\%$ and $2.9 \pm 0.22\%$, respectively. Cumulatively, the desired edit-to-indel ratios induced by mini-Sdd9-QBEmax were 3.26, 3.99, 2.02, 9.91 or 1.99-fold higher than that of mini-Sdd9-BE4max (Fig. 2d). Importantly, average product purities at the target cytosine base for stop codon creation by mini-Sdd9-QBEmax and mini-Sdd9-BE4max were 99.7% versus 95.7% for *PD-1*, 99.7% versus 97.7% for *CISH*, 99.7% versus 96.5% for *Fas*, 99.8% versus 98.0% for *TGFBR2* and 99.5% versus 95.8% for *TRAC* (Fig. 2e). This increase in product purity minimizes the formation of missense mutations from imprecise C-to-G or C-to-A edits (Fig. 2f). When analyzed individually, mini-Sdd9-QBEmax induced lower indels at 47 of the 51 target sites and higher product purities at 45 of the 51 target sites (Extended Data Figs. 3–7).

For each gene, we identified one ideal guide and next edited *PD-1*, *CISH*, *Fas*, *TGFBR2* and *TRAC* simultaneously for multiplex gene knockout. We transformed plasmids for each of the five sgRNAs together with QBEmax or BE4max editors into HEK293T cells. We observed

that mini-Sdd9-QBEmax exhibited comparable or higher editing compared to mini-Sdd9-BE4max across all five sites in HEK293T cells in the absence of any selection pressure (Fig. 2g). Notably, mini-Sdd9-QBEmax achieved lower indel formations (Fig. 2h) and exhibited superior product purity (Fig. 2i) at all five genes. To validate that all five base edits occurred in a single cell, we sequenced 48 and 112 QBEmax-transfected single-cell colonies arising from unsorted or sorted cell populations, respectively. We found that 21 (43.8%) and 100 (89.3%) cell colonies exhibited all five genes edited, respectively, demonstrating successful multiplex base editing by QBEmax in a single cell (Fig. 2j). To further evaluate the potential of QBEmax, we co-electroporated QBEmax and all five sgRNA plasmids into an immortal Jurkat T cell line. In these T cells, QBEmax also exhibited superior editing efficiencies, improved product purities and decreased indels, which is similar to its performance in HEK293T cells and further supports the versatility of QBEmax for safe and robust base editing in clinical applications (Extended Data Fig. 8).

Because DNA off-targets are a major concern for base editing therapeutic applications, we next evaluated Cas-independent DNA off-target effects of mini-Sdd9-QBEmax using the orthogonal R-loop assay^{23–25}. We cotransfected a dead-SaCas9 (dSaCas9) and sgRNA to induce the formation of an orthogonal R-loop simultaneously with the multiplexed gene knockout strategy (Fig. 2k). We evaluated five orthogonal sites and deep sequencing at each orthogonal R-loop showed that mini-Sdd9-QBEmax induced lower Cas-independent off-target editing at all five R-loop sites compared to that of mini-Sdd9-BE4max (Fig. 2l). We next evaluated the RNA off-target effects of mini-Sdd9-QBEmax. We conducted whole transcriptome sequencing and analyzed the number of C-to-U variants in QBEmax, BE4max and nCas9 (D10A) treated samples together with a sgRNA plasmid targeting *CISH*. We found that QBEmax induced substantially lower RNA off-target edits on transcriptome-wide RNA transcripts compared to that of the BE4max without compromising DNA on-target editing (Fig. 2m and Supplementary Fig. 5). The robust desired editing efficiencies, minimized indels, high product purities and decreased DNA and RNA off-target effects portray QBEmax as an ideal base editor for multiplex gene knockout applications.

We next sought to probe the molecular basis by which mini-Sdd9-QBEmax embodies its desired properties. We performed molecular dynamic (MD) simulation analyses based on the AlphaFold3-predicted ternary structures of mini-Sdd9-QBEmax or BE4max with a sgRNA and target DNA (Fig. 1d). With these models, all-atom MD simulations of approximately 300 ns were performed (Supplementary Fig. 6a,b). We first investigated the conformational stability of these two systems by projecting their free energy landscapes onto corresponding root mean square deviation (RMSD) and radius of gyration (Rg) components. We found that both the RMSD and Rg of mini-Sdd9-QBEmax were lower than that of mini-Sdd9-BE4max, and only one single stable energy state was observed (Fig. 3a,b and Supplementary Fig. 6b). This suggests that the QBEmax architecture better treats the deaminase and Cas protein as one complex so that each domain is oriented compactly within itself. At the minimum energy state, while the deaminase was predicted to be associated with the nontarget strand in both systems (Supplementary Fig. 6c,d), the RMSD of mini-Sdd9-QBEmax system was lower throughout the 300 ns MD process (Supplementary Fig. 6b). When analyzing individual amino acids, we observed that the root mean square fluctuation (RMSF) of the linkers connecting the deaminase to the Cas protein were lower in the QBEmax system (Fig. 3c,d) compared to the BE4max architecture. We speculate a more compact QBEmax architecture that limits the deaminase from sporadically swinging in space, thereby contributing to lower Cas-independent DNA off-target editing, lower indel formation and higher product purity.

During cytosine base editing, intermediate uracil cleavage by endogenous uracil DNA glycosylase (UNG) drives the formation of indels and imprecise C-to-G or C-to-A edits^{1,26}. We envisioned that an

ideal base editor adopts a compact and protective conformation for the exposed R-loop so that the intermediate uracil base is not excised before cellular mismatch repair resolving a permanent C-to-T conversion. To evaluate R-loop exposure, we performed solvent accessibility analyses using a 1.0 nm probe and found that the solvent-accessible surface area (SASA) was increased for C3 and substantially increased for C7 and C8 in the mini-Sdd9-BE4max compared to the mini-Sdd9-QBEmax architectures (Fig. 3e,f), suggesting that these residues are accessible by UNG and ultimately form indels and byproducts. We also speculated that the distance of the UGI to the ssDNA target bases may affect product purity. We evaluated the position and distance of the two UGI domains to the ssDNA target bases in the QBEmax, BE4max or individual CP-1031 and inlaid-1244 CBEs based on molecular dynamic modeling data. We observed that indeed both UGI domains in QBEmax exhibited a relatively shorter distance to the target bases in the ssDNA R-loop region, which suggests an inverse relationship between UGI positioning to product purities and indel formation, as others previously have also identified⁵ (Supplementary Fig. 6e).

Based on these results, we propose a model for QBEmax base editing. In cytosine base editing using a canonical BE4max architecture, indel formation and impure C-to-G or C-to-A base edits arise from uracil excision and abasic site formation. Because QBEmax exhibits a more compact architecture, limits deaminase swinging and shields the Cas9-induced R-loop, base editing intermediates are protected from cellular UNG excision before Cas9 detaching from the target DNA and subsequent mismatch repair. Therefore, a protective and compact base editor conformation reduces unintended effects driven by DNA repair processes and further promotes desired base editing events (Extended Data Fig. 9).

Taken together, we designed and identified a base editor architecture, QBEmax, which achieves high efficiency on-target editing while decreasing low indel formation, exhibits high edit product purities and minimizes DNA off-targets. The development of QBEmax serves as a promising base editor architecture for developing more efficient and precise base edits toward the use of base editing in multiplex therapeutic applications such as CAR-T immunotherapies. The efficient delivery of QBEmax in vivo will further expand on its use and analysis of desired base editing properties. Advances in protein prediction and MD further help shed light on the molecular basis of genome editors and would greatly aid in future developments of new editing technologies.

Online content

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Methods

Plasmid construction

The construct fragments were PCR amplified using 2× Phanta Max Master Mix (Vazyme Biotech) and 2× KOD one Master Mix (TOYOBO Life Sciences) and cloned into the pCMV²⁴ backbone, using Uniclone One Step Seamless Cloning Kit (Genesand). Plasmids for HEK293T cell line transfection were extracted and purified using EndoFree Plasmid Kits (Qiagen) or FastPure EndoFree Plasmid Mini Kit DC203 (Vazyme Biotech). The amino acid sequences of the representative vectors used are provided in the Supplementary Note. Primers used for amplicon sequencing were synthesized by the Beijing Genomics Institute and are listed in Supplementary Table 2.

Cell culture and transfection

HEK293T, A549, HeLa and HCT116 cells are cultured in DMEM (Gibco) supplemented with 10% (vol/vol) FBS (Gibco) and 1% (vol/vol) penicillin–streptomycin (Gibco); Jurkat, Clone E6-1 cells are cultured in Roswell Park Memorial Institute (RPMI, Gibco) 1640 medium supplemented with 10% (vol/vol) FBS (Gibco) in a humidified incubator at 37 °C with 5% CO₂. All the cells were routinely tested for Mycoplasma contamination with a mycoplasma detection kit (TransGen Biotech). For HEK293T cells transfection, 6 × 10⁴ cells per well were seeded into 48-well poly-D-lysine-coated plates (Corning) in the absence of antibiotic. After 16–24 h, plasmids were transfected into the cells; for single target base editing, cells were transfected with 1 µl jetPRIME transfection reagent (Polyplus), 375 ng editor plus 125 ng sgRNA plasmids per well, for simultaneous editing of the five CAR-T relevant genes and the R-loop assays, 200 ng of editor plasmid and 66 ng of sgRNA plasmid for each gene was transfected, together with 200 ng of dSaCas9 plasmid and 66 ng of an orthogonal R-loop inducing sgRNA plasmid, at a 60–80% cell confluency. Cells were washed with PBS and followed by DNA extraction 72 h after transfection. To isolate single-cell colonies, transfected cells with or without sorting were diluted and plated into 96-well plates at a density of 0.9 cells per well. The cell colonies were cultured for 2 to 3 weeks and then transferred into 48-well plates to grow for another 5 days before cell lysis, DNA extraction and sequencing of all five targeted genomic loci. Effective editing in a single-cell colony for any individual gene is benchmarked as having an editing efficiency surpassing 50% at the targeted base when analyzed by CRISPResso2 (ref. 27). For the sorted cell population, sgRNA plasmids and a mini-Sdd9-QBEmax-P2A-mScarlet plasmid were transfected and the top 5% of mScarlet-positive cells were collected and diluted into single-cell colonies. For A549, HeLa and HCT116 cells, 5 × 10⁴, 2 × 10⁵ or 4 × 10⁵ cells were seeded into 12-well poly-D-lysine-coated plates (Corning), respectively, in the absence of antibiotic. After 16–24 h, 375 ng of mini-Sdd9-BE4max-P2A-mScarlet or mini-Sdd9-QBEmax-P2A-mScarlet plasmids were transfected into cells together with 125 ng sgRNA plasmid while using 2 µl of jetPRIME transfection reagent (Polyplus). After 48 h, cells were resuspended for fluorescence-activated cell sorting. For Jurkat cells, 1,000 ng of mini-Sdd9-BE4max-P2A-mScarlet or mini-Sdd9-QBEmax-P2A-mScarlet editor plasmids and 250 ng of sgRNA plasmid for each of the five genes were cotransfected into 4 × 10⁵ cells using Entranster-E (Engreen Biosystem) and the 4D-Nucleofector (Lonza Biosciences) with program DS167. mScarlet-positive cells were collected for DNA extraction.

Fluorescence-activated cell sorting

Cells were collected and resuspended in a culture medium supplemented with 2% FBS in a 0.5 ml volume. Cells were sorted on a FACSAria III (BD Biosciences) cytometer after gating for the singlet-cell population by the mScarlet signal. For RNA off-target analysis, both the fluorescence-positive and negative cells were collected for further paired analysis. A representative flow cytometry gating strategy can be found in Supplementary Fig. 7.

Genomic DNA isolation from mammalian cell culture

Genomic DNA extraction was performed by the addition of 100 µl freshly prepared lysis buffer (10 mM Tris–HCl (pH8.0), 0.05% SDS and 25 µg ml^{−1} proteinase K (Thermo Fisher Scientific) directly into the 48-well culture plate after cells were washed once with 1× Dulbecco's PBS (Thermo Fisher Scientific). The mixture was incubated at 37 °C for 60 min and then treated at 80 °C in the thermocycler for 20 min.

Transcriptome-wide RNA off-target analysis

To profile the transcriptome-wide RNA off-target effects of the QBEmax and BE4max, 2 × 10⁵ HEK293T cells were seeded into 12-well poly-D-lysine-coated plates (Corning) in the absence of antibiotics. After 24 h, 750 ng of mini-Sdd9-BE4max-P2A-mScarlet, mini-Sdd9-QBEmax-P2A-mScarlet or Cas9(D10A)-P2A-mScarlet plasmids were transfected into cells together with 250 ng sgRNA plasmid targeting the *CISH*-site 1 using 2 µl of jetPRIME transfection reagent (Polyplus). After 48 h, cells were collected for cell sorting. For each treatment, both the mScarlet-positive and negative cells were collected for paired analyses. RNA samples were sequenced using an MGI T7 (2 × 150 PE) platform at the JMDNA, at a depth of ~45 million reads per sample. Raw reads were cleaned by fastp²⁸. The reads were mapped to the human reference genome (hg38) by Hisat2 (ref. 29) software (version 2.2.1). After removing duplication, variants were identified using strelka³⁰ (version 2.9.10). Finally, C-to-U edits in the transcribed strand were considered for downstream analyses.

Amplicon deep sequencing and data analysis

Two rounds of PCR were used to amplify a DNA fragment spanning the target site. In the first round PCR, the target region was amplified from genomic DNA with site-specific primers using the DNA template. In the second round, both forward and reverse barcodes were added to the ends of the PCR products for library construction. Equal amounts of PCR product were pooled and purified with a FastPure Gel DNA Extraction Mini Kit (Vazyme Biotech, Inc.) and quantified with a Qubit 4 (Thermo Fisher Scientific). The purified products were sequenced using the Illumina Miseq platform or the BGI G99 platform, and the sequences around the target regions were examined to analyze the editing outcomes. Sequences of the NGS primers and the corresponding amplicons are listed in Supplementary Table 2. Analysis of the base editing outcomes was performed as described previously³¹.

MD simulations and analyses

Initial protein structure models of mini-Sdd9-BE4max and mini-Sdd9-QBEmax ternary complex were predicted using AlphaFold3 (ref. 14), with the optimal model_0 selected as the starting structure for MD simulations. MD simulations were conducted using GROMACS³² (version 2023.1, CUDA) with the Amber99BSC1 force field and the TIP3P water model^{33,34}. In each simulation system, the complex was positioned at the center of a cubic box with a distance of 15 Å maintained between the surface and the edges of the box. Na⁺ and Cl[−] ions were added to the system to achieve charge neutrality with an ion concentration set at 0.15 M. During the simulation, the particle mesh Ewald method³⁵ was employed to calculate electrostatic interactions and a cut-off distance of 14 Å was used for analysis of short-range electrostatic and van der Waals forces. The LINC algorithm³⁶ was applied to constrain bonds involving hydrogen atoms, and periodic boundary conditions were specified in all three dimensions. To optimize the system, energy minimization was performed using the steepest descent method, continuing until a maximum of 2,500 steps was reached or the maximum force <1,000 kJ mol^{−1} nm^{−1}. Next, 100 ps of NVT equilibration and 100 ps of NPT equilibration were conducted. During the equilibration process, the V-rescale temperature and Parrinell–Rahman pressure coupling methods were employed, with the system's temperature and pressure set to 310 K and 1.0 bar, respectively. After system equilibration, a 300 ns MD simulation was performed.

RMSD and RMSF data were generated using the `gmx rmsd` and `gmx rmsf` commands in GROMACS. The SASA data were obtained through the `gmx sasa` command, with a probe diameter of 1.0 nm, and snapshots of the SASA state were collected every 1 ns within the stable region. All data were statistically analyzed and visualized using R packages `ggplot2` (3.5.1) and `tidyverse` (2.0.0). The free energy landscape data were generated using the `gmx sham` command with the parameter `-tsham` set to 310 K. The data were plotted along the two projection components of RMSD and Rg. Visualization of this data was performed using the `matplotlib` (3.8.2) package in Python³⁷.

Statistical analysis

GraphPad Prism 9 software was used to analyze the data. All numerical values are presented as means $\pm$ s.e.m. Significant differences between controls and treatments were tested using the Student's *t*-test or one-way ANOVA Tukey's multiple comparisons. $P < 0.05$ was considered statistically significant, and $P < 0.01$ was considered statistically extremely significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The deep amplicon sequencing data have been deposited in the NCBI BioProject database under accession code [PRJNA1147008](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1147008) (ref. 38). Plasmids encoding QBEmax and QBEn are available at Addgene. All other data are available in the main paper or Supplementary Information.

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Author contributions

K.T.Z. and J.H. conceptualized the study. M.G., J.H., M.H., G.L. and Quan Gao assembled the vectors and conducted experiments in HEK293T cells. M.H., L.G. and G.L. performed amplicon sequencing. J.H. and M.G. collected and analyzed amplicon sequencing data. Qiang Gao and H.J. performed the MD analysis. Qiang Gao and Z.W. wrote scripts and processed the raw amplicon sequencing data. J.H., M.G., M.H. and K.T.Z. prepared the figures. J.H. and K.T.Z. wrote the manuscript with input from all authors.

Competing interests

The authors have submitted a patent application based on the results reported in this paper. K.T.Z. is the founder and holds equity at Qi Biodesign. J.H., M.G., Qiang Gao, H.J., M.H., Z.W., L.G., G.L. and Quan Gao are employees of Qi Biodesign.

Additional information

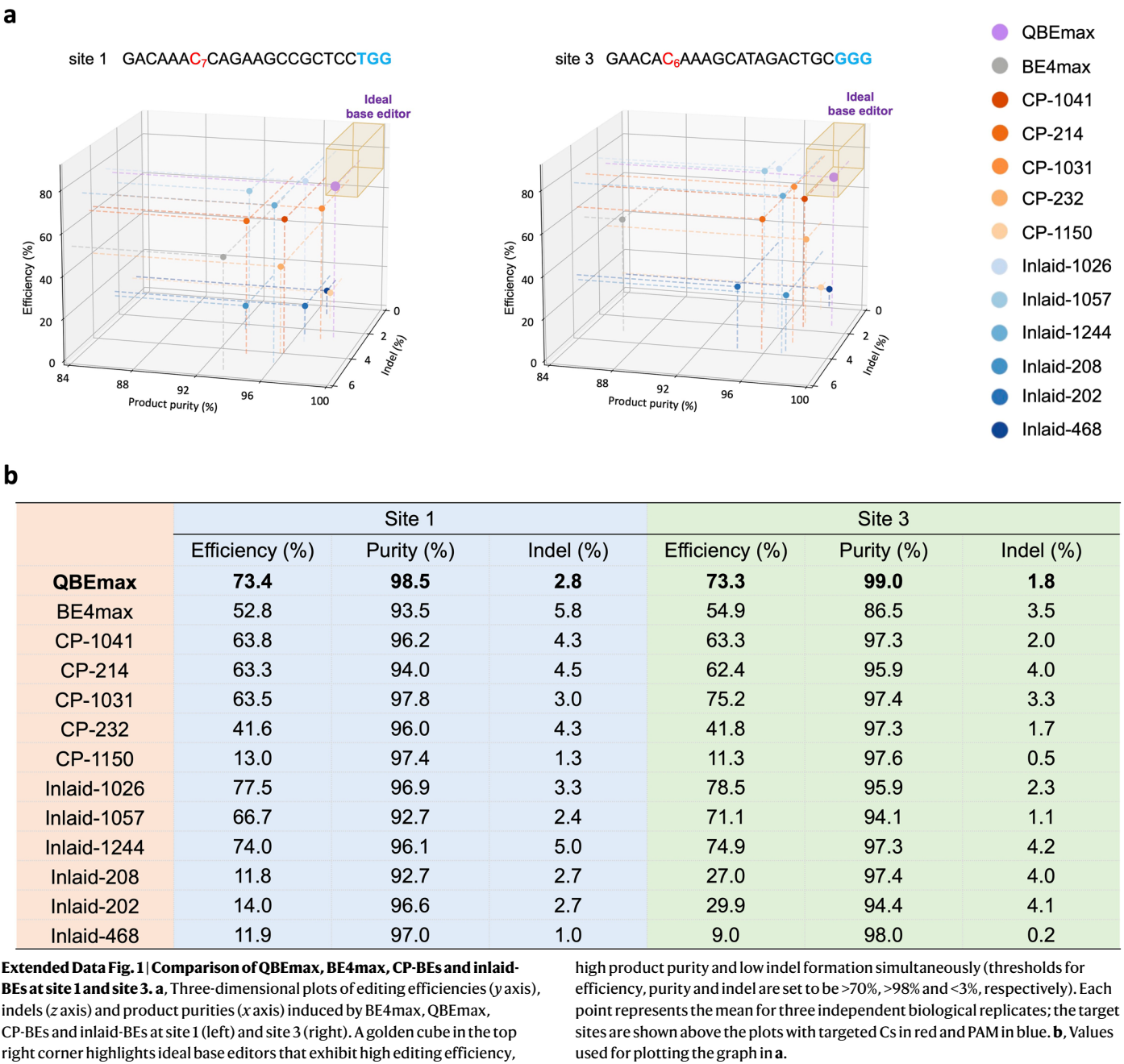
Extended data is available for this paper at <https://doi.org/10.1038/s41587-025-02641-9>.

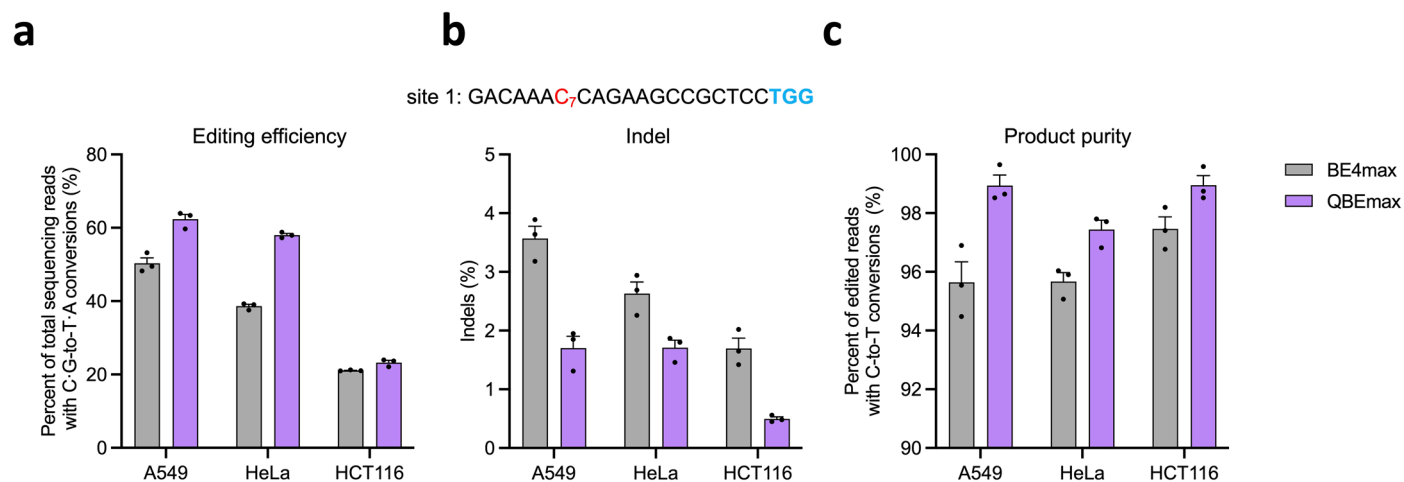
Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41587-025-02641-9>.

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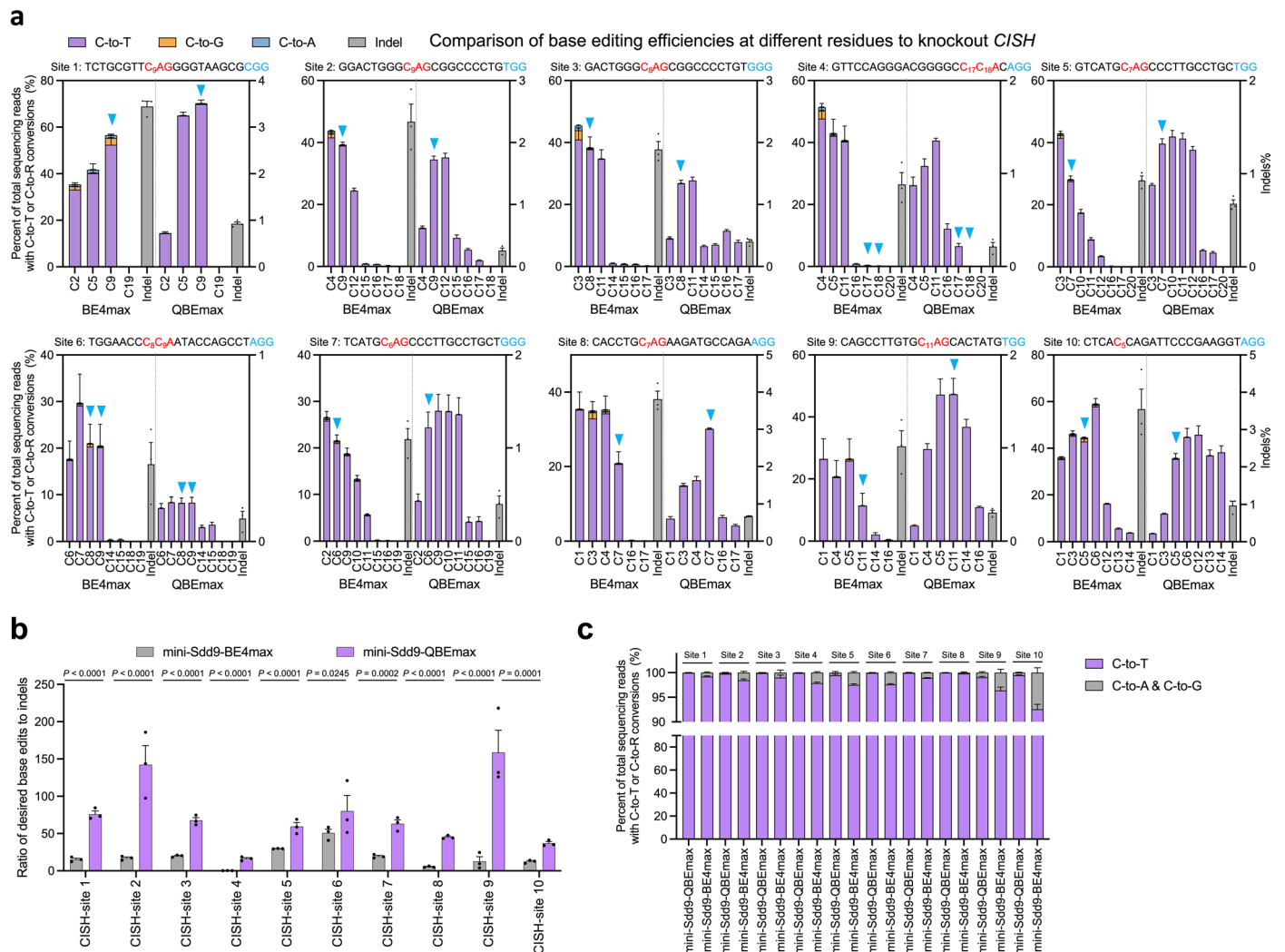
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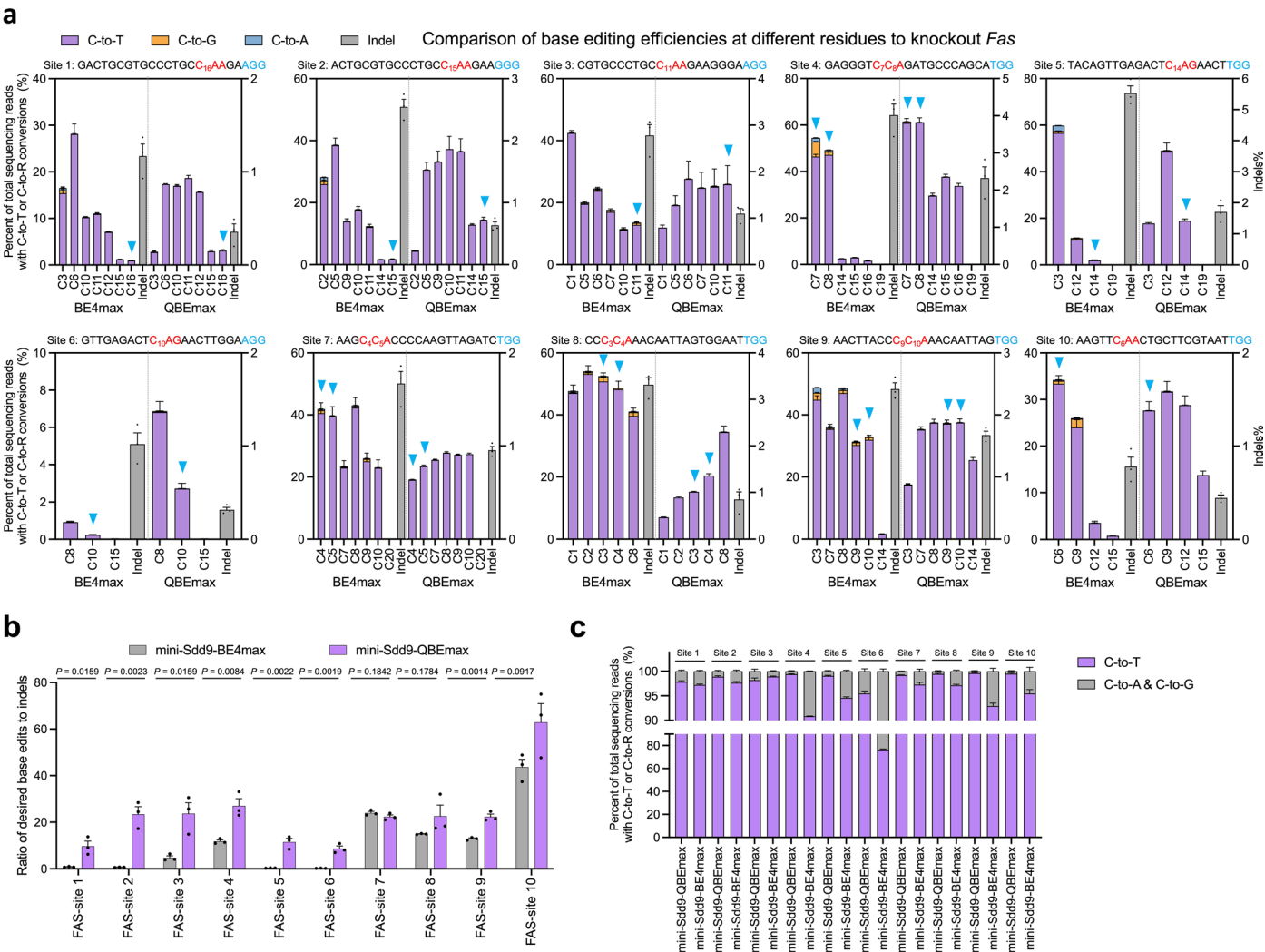
Extended Data Fig. 2 | Performance of QBEmax in other mammalian cell lines.
a–c. Desired editing efficiencies (**a**), indels (**b**), and percent of edited reads (**c**) with C-to-T conversions induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax

editors at site 1 with C₇ in red used to calculate the efficiency and product purity. Values and error bars represent the means and s.e.m. for three independent biological replicates.



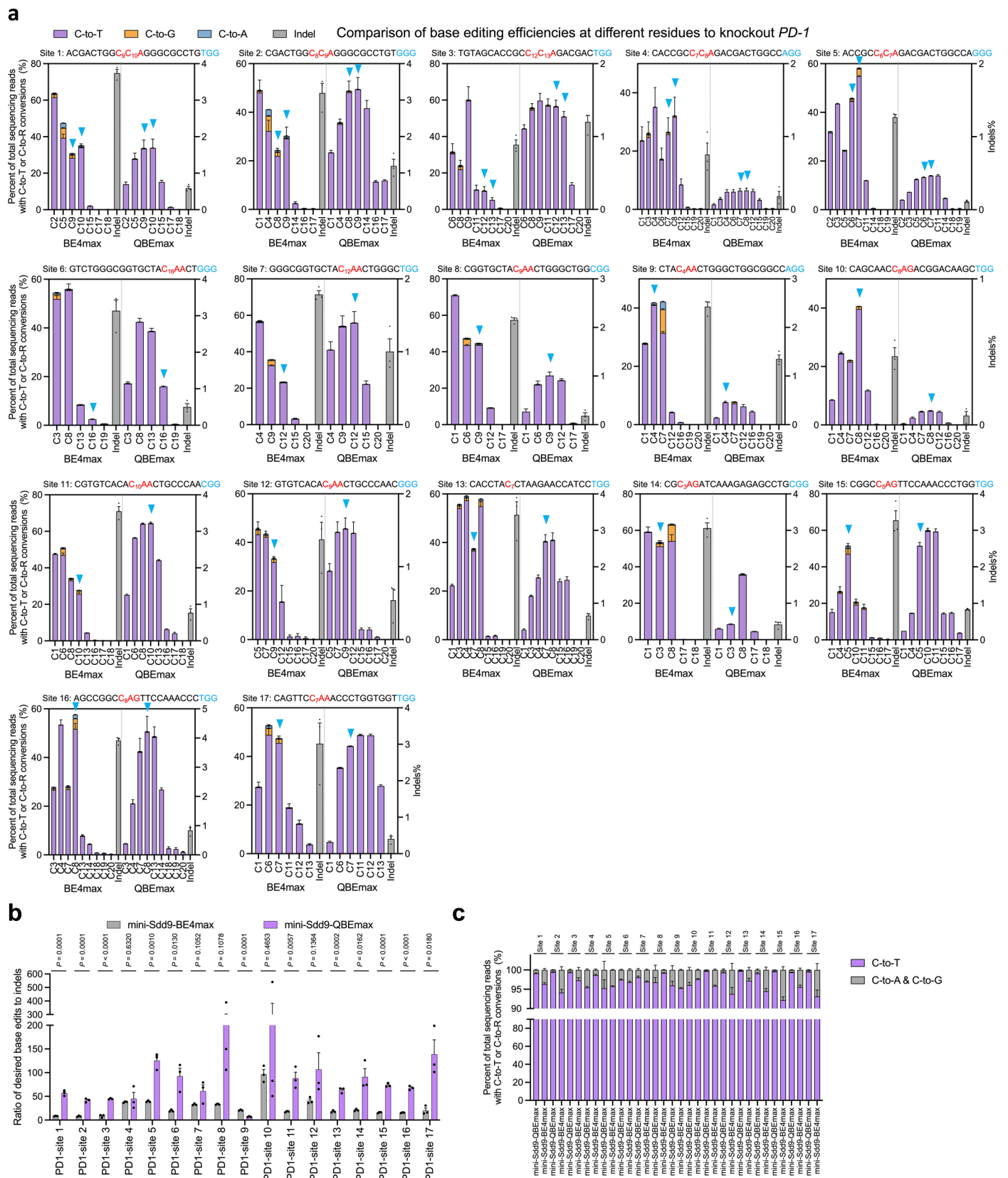
Extended Data Fig. 3 | Evaluation of BE4max and QBEmax for base editing-mediated knockout of *CISH*. **a**, Frequencies of C-to-T or C-to-R conversions (left y axis) and indels (right y axis) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax for ten sites targeting *CISH* knockout. **b**, Ratio of base edits to indels induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax; significances

are indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBEmax by exact *P* value using two-tailed Student's *t* test. **c**, Percent of edited reads with C-to-T or C-to-R conversions at the target sites indicated. Values and error bars represent the means and s.e.m. for three independent biological replicates.



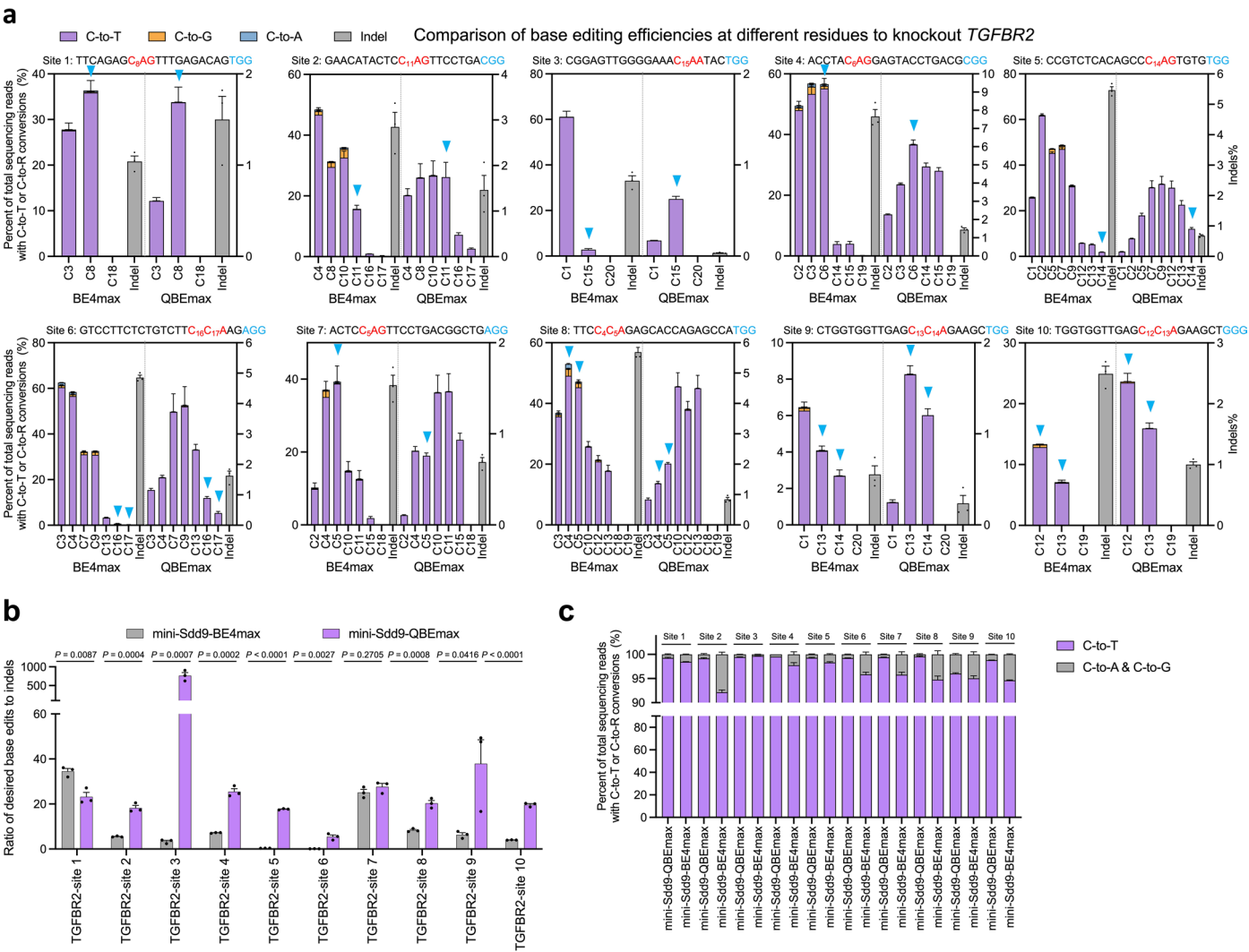
Extended Data Fig. 4 | Evaluation of BE4max and QBE4max for base editing-mediated knockout of *Fas*. **a**, Frequencies of C-to-T or C-to-R conversions (left y-axis) and indels (right y-axis) induced by mini-Sdd9-BE4max and mini-Sdd9-QBE4max for ten sites targeting *Fas* knockout. **b**, Ratio of base edits to indels induced by mini-Sdd9-BE4max and mini-Sdd9-QBE4max; significances

are indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBE4max by exact *P* value using two-tailed Student's *t* test. **c**, Percent of edited reads with C-to-T or C-to-R conversions at the target sites indicated. Values and error bars represent the means and s.e.m. for three independent biological replicates.



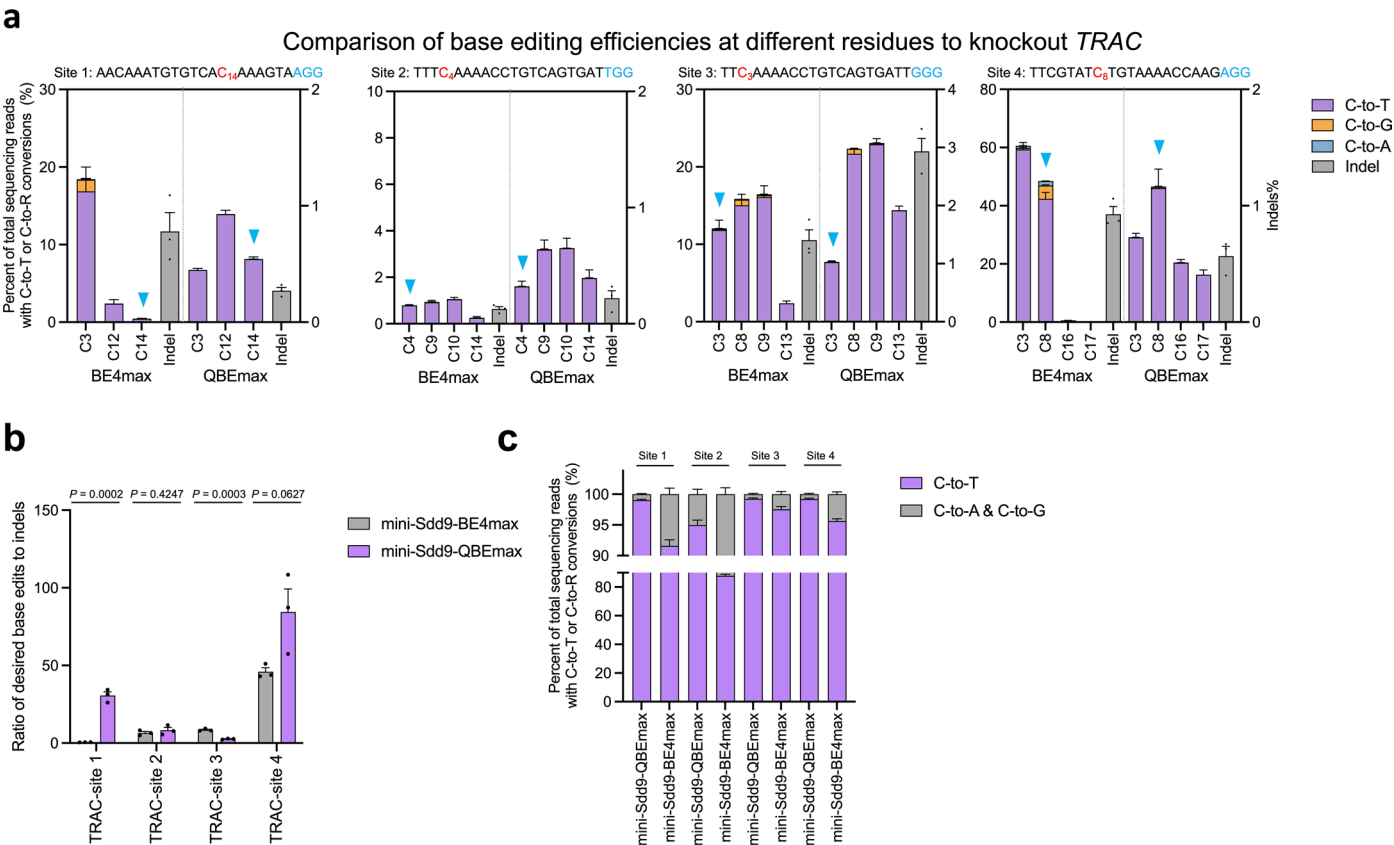
Extended Data Fig. 5 | Evaluation of BE4max and QBEmax for base editing-mediated knockout of *PD-1*. **a**, Frequencies of C-to-T or C-to-R conversions (left y-axis) and indels (right y-axis) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax for 17 sites targeting *PD-1* knockout. **b**, Ratio of base edits to indels induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax; significances

are indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBEmax by exact *P* value using two-tailed Student's *t* test. **c**, Percent of edited reads with C-to-T or C-to-R conversions at the target sites indicated. Values and error bars represent the means and s.e.m. for three independent biological replicates.

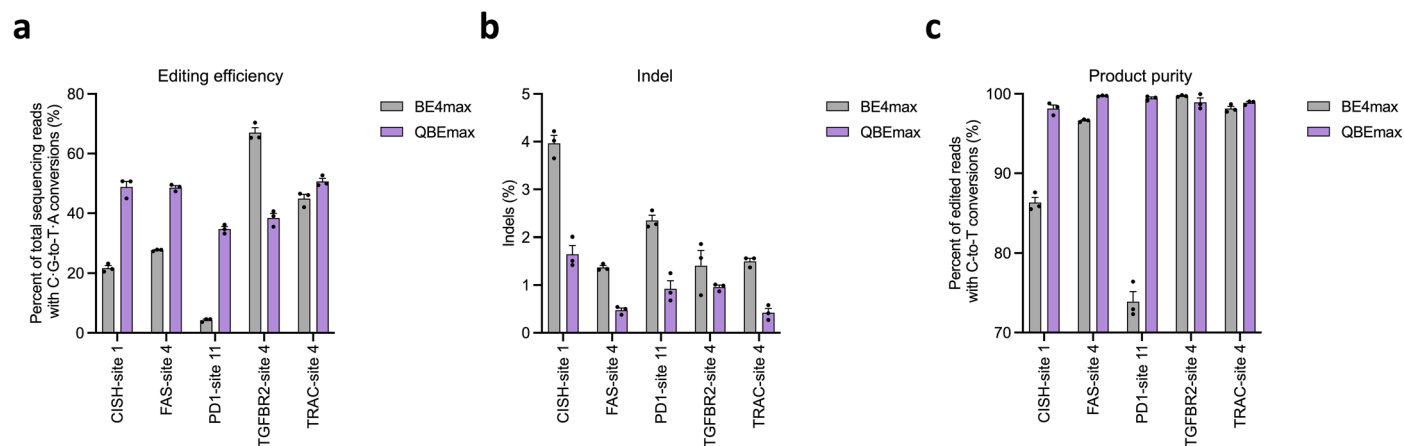


Extended Data Fig. 6 | Evaluation of BE4max and QBEmax for base editing-mediated knockout of *TGFR2*. **a**, Frequencies of C-to-T or C-to-R conversions (left y axis) and indels (right y axis) induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax for ten sites targeting *TGFR2* knockout. **b**, Ratio of base edits to indels induced by mini-Sdd9-BE4max and mini-Sdd9-QBEmax; significances are

indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBEmax by exact *P* value using two-tailed Student's *t* test. **c**, Percent of edited reads with C-to-T or C-to-R conversions at the target sites indicated. Values and error bars represent the means and s.e.m. for three independent biological replicates.

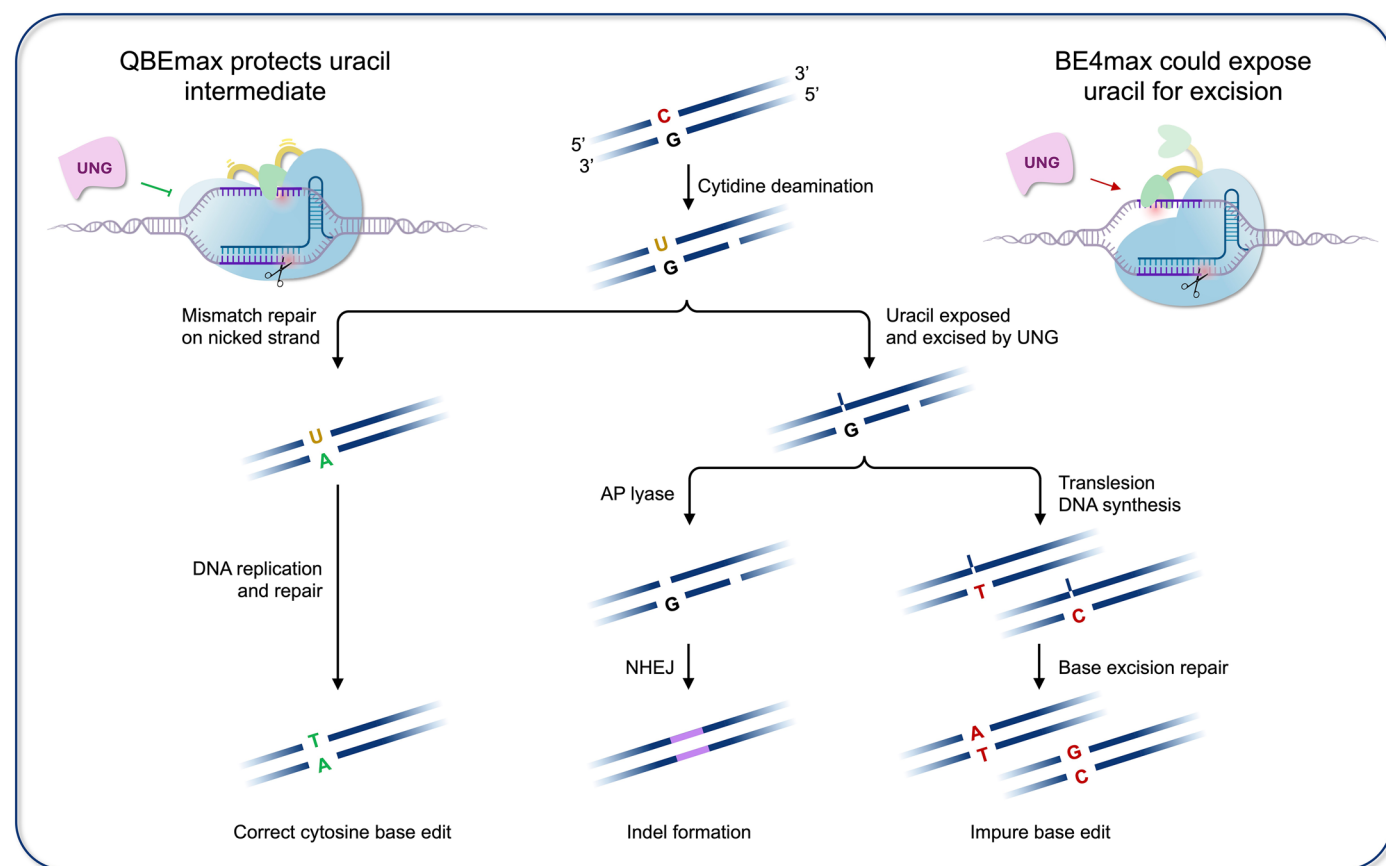


Extended Data Fig. 7 | Evaluation of BE4max and QBEmax for base editing-mediated knockout of *TRAC*. **a**, Frequencies of C-to-T or C-to-R conversions (plot on the left y axis) and indels (plot on the right y axis) induced by the mini-Sdd9-BE4max and mini-Sdd9-QBEmax for four sites targeting *TRAC* knockout. **b**, Ratio of base edits to indels induced by the mini-Sdd9-BE4max and the mini-Sdd9-QBEmax; significances are indicated between the mini-Sdd9-BE4max and mini-Sdd9-QBEmax by exact *P* value using two-tailed Student's *t* test. **c**, Percent of edited reads with C-to-T or C-to-R conversions at the target sites indicated. Values and error bars represent the means and s.e.m. for three independent biological replicates.



Extended Data Fig. 8 | Multiplex base editing with QBE4max and BE4max in immortal Jurkat cells. a–c. Desired editing efficiencies (a), indels (b) and percent of edited reads (c) with C-to-T conversions at target genes induced by

mini-Sdd9-BE4max and mini-Sdd9-QBE4max editors during multiplexed base editing of five genes; values and error bars represent the means and s.e.m. for three independent biological replicates.



Extended Data Fig. 9 | Model showing base editing outcomes for BE4max and QBE4max. UNG, uracil DNA glycosylase; AP lyase, apurinic or apyrimidinic site lyase; NHEJ, nonhomologous end joining.

Reporting Summary

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Only common tests should be described solely by name; describe more complex techniques in the Methods section.
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Give P values as exact values whenever suitable.
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Software and code

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Data collection

Illumina Miseq and BGI G99 platforms were used to collect the amplicon deep sequencing data. RNA samples were sequenced using an MG T7 (2 × 150 PE) platform. Cells were sorted on a FACS Aria III (BD Biosciences) cytometer.

Data analysis

Amplicon sequencing data was analyzed using the published code as previously described and noted in the methods. Graphpad prism 9 was used to analyze the data.

Flow cytometry data was analyzed using the BD FACSDiva™ Software (Version 9.7).

Raw RNA reads were cleaned by fastp. The reads were mapped to the human reference genome (hg38) by Hisat2 software (Version 2.2.1). After removing duplication, variants were identified using strelka29 (Version 2.9.10).

Initial protein structure models of mini-Sdd9-BE4max and mini-Sdd9-QBE4max ternary complex were predicted using AlphaFold3, with the optimal model_0, selected as the starting structure for MD simulations. MD simulations were conducted using GROMACS (version 2023.1, CUDA) with the Amber99BSC1 force field and the TIP3P water model. In each simulation system, the complex was positioned at the center of a cubic box with a distance of 15 Å maintained between the surface and the edges of the box. Na⁺ and Cl⁻ ions were added to the system to achieve charge neutrality with an ion concentration set at 0.15 M. During the simulation, the Particle Mesh Ewald (PME) method was employed to calculate electrostatic interactions and a cutoff distance of 14 Å was utilized for analysis of short-range electrostatic and van der Waals forces. The LINCS algorithm was applied to constrain bonds involving hydrogen atoms, and periodic boundary conditions (PBC) were specified in all three dimensions. To optimize the system, energy minimization was performed using the steepest descent method, continuing until a maximum of 2500 steps was reached or the maximum force 1000 kJ·mol⁻¹·nm⁻¹. Next, 100 ps of NVT equilibration and 100 ps of NPT equilibration were conducted. During the equilibration process, the V-rescale temperature and Parrinell-Rahman pressure coupling methods were employed, with the system's temperature and pressure set to 310 K and 1.0 bar, respectively. After system equilibration, a 300 ns MD simulation was performed.

RMSD and RMSF data were generated using the gmx rmsd and gmx rmsf commands in GROMACS (version 2023.1, CUDA). The SASA data

were obtained through the gmx sasa command, with a probe diameter of 1.0 nm, and snapshots of the SASA state were collected every 1 ns within the stable region. All data were statistically analyzed and visualized using R packages ggplot2 (version 3.5.1) and tidyverse (version 2.0.0). The free energy landscape data were generated using the gmx sham command with the parameter -tsham set to 310 K. The data were plotted along the two projection components of RMSD and Rg. Visualization of this data was performed using the matplotlib (version 3.8.2) package in Python.

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The human reference genome hg38 was used for data analysis in this study. The authors declare that all data supporting the findings of this study are available in the article or its Supplementary Information files. The deep amplicon sequencing data is deposited in NCBI under BioProject Accession Code PRJNA1147008.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The experiments in HEK293T, A549, HeLa, HCT116, and Jurkat cells were performed with at least three independent biological replicates for all data with statistical analyses. Cell transfections were performed as described in the methods.
Data exclusions	No data exclusion.
Replication	All attempts for replication were successful. For the experiments in HEK293T cells, a minimum of three independent experiments were included.
Randomization	All cells were separately passaged and transfected to ensure biological randomization.
Blinding	Not applicable. As samples were processed identically through standard and in some cases automated procedures (DNA sequencing, transfection, DNA isolation) that should not bias outcomes.

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Materials & experimental systems

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☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	The HEK293T, A549, HeLa, and HCT116 cells were obtained from ATCC. The Jurkat, Clone E6-1 cells were kindly provided by Cell Bank, Chinese Academy of Sciences.
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Authentication

The cells with validated by the company or academy and authenticated by STR profiling.

Mycoplasma contamination

Validated by the company and tested in our studies to ensure no mycoplasma contamination

Commonly misidentified lines
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No commonly misidentified lines were used in this study.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
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- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

All cells were cultured and transfected as described in the methods.

Instrument

Cells were sorted on a FACSARIA III (BD Biosciences) cytometer.

Software

BD FACSDiva™ Software (Version 9.7)

Cell population abundance

For A549, HeLa and HCT116 cells, 50k, 200k or 400k cells were seeded and cultured for two days. For Jurkat cells, 400k cells were used for electroporation and cultured for two days.

Gating strategy

Negative control (untreated) and fluorophore-positive cells were used to establish gates for each cell type. Gates were drawn to collect cells expressing either fluorophore.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.